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Something changed in the last four years, and every defence ministry on earth is still catching up with it. Mass, budget and pedigree stopped deciding outcomes. A two-thousand-dollar quadcopter, built by volunteers, now does the work of a thirteen-million-dollar missile. The advantage has moved to whoever can think, build and field fastest, and it is not moving back.

For Singapore that is the best news in a generation and the most dangerous. We cannot win a war of depth — we have fifty kilometres and no rear area. We cannot win a war of stockpiles, because what we would need most is exactly what nobody will sell us at the moment we need it. What we can win is a war of adaptation, and that contest is decided by engineers.

Which is why you are here. The four problems that follow are not exercises. They are unsolved by anyone, and they are the ones that decide whether this island holds. Nobody expects you to finish one in a weekend. We expect you to see it clearly, come at it from an angle nobody has tried, and show us that it works.

Forty-eight hours. Pick your problem and begin.

§1

THE FOUR CHALLENGES

01

GREEN CORRIDOR

Ideate a machine that takes a dangerous job off a person in Singapore's terrain — jungle, mud, coast or tidal ground. What it does is your decision. Showcase a proof of concept that makes the mechanism believable: a bench rig, a simulation, or one subsystem demonstrated. See document 01 for more details.

02

CREATIVE SENSING

Ideate creative ways to detect something that emits no radio, returns almost nothing to radar, and sits at the temperature of the air around it. Showcase a demo that reports how often it cries wolf against a real background — not only how far it can see. See document 02 for more details.

03

INTERCEPTORS

Ideate one layer of an autonomous interceptor swarm that needs no pilot — the drone itself, the terminal guidance, the swarm deployment logic, or the patrol and recharge cycle that keeps it airborne. Build it in software and showcase it working, measured against a stated baseline. See document 03 for more details.

04

ONE PICTURE, MANY EYES

Ideate a way to fuse sources nobody has joined — sensors, open data, and things that were never meant to be sensors — into usable warning, either that a conflict is forming or of what is on our approaches now. Showcase a software demo that turns many disagreeing inputs into one picture somebody can act on. See document 04 for more details.

§2

SUBMISSION

2.1

VIDEO DEMONSTRATION

The prototype working. Thirty seconds maximum. If the working part is in simulation, record the simulation.

2.2

PRESENTATION DECK

Three minutes, spoken to rather than read from.

2.3

TECHNICAL ARCHIVE

One ZIP with everything needed to verify the work: source code, CAD and model files, simulation configurations, datasets, test outputs, bench measurements.

2.4

README

At the top of the archive. What each file is, and how to run what can be run.

The selection of approach rests with the team. The problem is specified here; the solution deliberately is not. What is assessed is creativity and the demonstration of technical feasibility. All the best.